

White Holes as Superconducting Phase Reversers

The Complete Theory of Black Hole–White Hole Symmetry in the Harmonic Framework

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Abstract

Black holes and white holes are traditionally viewed as time-reversed counterparts in general relativity. This paper extends the Harmonic Framework's superconductor model of black holes to include white holes, demonstrating that both are manifestations of the same superconducting phase—just reversed in time.

I show that:

1. **Black holes are superconducting phase converters.** They convert T_1 (matter) into T_2 (antimatter) at 0° phase, achieving the $m + m' = 0$ condition.
2. **White holes are superconducting phase reversers.** They convert T_2 (antimatter) back into T_1 (matter) at 0° phase, also achieving $m + m' = 0$.
3. **The black hole and white hole are the two mouths of an Einstein–Rosen bridge.** The bridge is a superconducting tunnel connecting T_1 and T_2 at 0° phase.
4. **Black hole jets are the white hole emission.** The T_2 standing wave is re-emitted along the spin axis as T_1 at 0° phase, forming coherent, phase-locked jets.
5. **The 27 Hz anchor governs both.** The characteristic frequency of the black hole–white hole system is $f_{\text{BH}} = 27 \text{ Hz} \times (M_{\text{sun}}/M) \times (\text{spin factor})$.
6. **The 90° phase lens effect applies to both.** Hawking radiation is the T_2 standing wave projected through a 90° phase shift. The white hole emission is the T_1 standing wave projected through a 90° phase shift.
7. **The information paradox is fully resolved.** Information is stored in the phase relationships of the T_2 standing wave inside the black hole and re-emitted through the white hole jets.
8. **The Einstein–Rosen bridge is a superconducting tunnel.** The throat connecting the black hole and white hole is a coherent T_1 – T_2 standing wave at 0° phase, allowing traversal without inertia.

The theory predicts that white holes should show the same 32-harmonic comb as black holes, that their jets should be phase-locked to the 27 Hz anchor, and that the Einstein–Rosen bridge should be

traversable when phase-locked.

Keywords: White holes, black holes, superconducting phase converters, Einstein–Rosen bridge, T_1 , T_2 , $m + m' = 0$, 27 Hz anchor, 32-harmonic comb, phase lens, jets, information paradox

1. Introduction: The Black Hole–White Hole Pair

1.1 The Standard View

General relativity predicts that black holes and white holes are time-reversed solutions of the Einstein field equations.

Black hole: A region from which nothing can escape.

White hole: A region from which nothing can enter—the time-reverse of a black hole.

The Einstein–Rosen bridge: A connection between a black hole and a white hole—a wormhole.

The problem: White holes are considered unstable and unphysical. They are thought to be mathematical curiosities, not real astrophysical objects.

1.2 The Harmonic Framework View

The Harmonic Framework provides a different interpretation:

1. **Black holes and white holes are real.** They are the two ends of a superconducting phase tunnel.
 2. **Both are superconductors.** They achieve $m + m' = 0$ at 0° phase.
 3. **They are phase converters and reversers.** Black holes convert $T_1 \rightarrow T_2$. White holes convert $T_2 \rightarrow T_1$.
 4. **The Einstein–Rosen bridge is a superconducting tunnel.** It connects T_1 and T_2 at 0° phase.
 5. **The jets are the white hole emission.** The T_2 standing wave is re-emitted as T_1 .
 6. **The information paradox is resolved.** Information flows through the bridge.
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2. The Harmonic Framework Recap

2.1 The Three Branches of Time

Branch	Time Dimension	n	Equation	Cutoff Frequency
Matter	T_1 (forward)	Yes (1, 2, 3, ..., 54.65)	$E = m \cdot c^2 \cdot (v/c)^{(n)}$	None (continuous)
Antimatter	T_2 (reverse)	No	$m_{\text{anti}} = \frac{(h \cdot f_{c2}/c^2)}{\times k}$	$f_{c2} = 27/230 = 0.1174 \text{ Hz}$
Dark Matter	T_3 (orthogonal)	No	$m_{\text{dm}} = \frac{(h \cdot f_{c3}/c^2)}{\times k}$	$f_{c3} = 27 \times (13/19)/230 = 0.08032 \text{ Hz}$

2.2 The Phase Relationships

Pair	Phase Difference	Interaction
$T_1 + T_2$ (superconductor)	0°	Mass cancellation ($m + m' = 0$)
$T_1 + T_2$ (annihilation)	180°	Mass-to-energy conversion
$T_1 + T_3$	90°	Inert
$T_2 + T_3$	180°	Annihilation

2.3 The 27 Hz Anchor

The Tau lattice is anchored at $f_0 = 27$ Hz.

2.4 The 230th Subharmonic

$$f_{c2} = 27 \text{ Hz} / 230 \approx 0.1174 \text{ Hz}$$

2.5 The -1° Phase Offset

$$\varphi_0 = -1^\circ = -\pi/180 \text{ radians}$$

3. Black Holes as Superconducting Phase Converters

3.1 The Mechanism

In a black hole:

1. Infalling matter (T_1) crosses the event horizon
2. It is split into T_1 and T_2 components
3. They are phase-locked at 0°
4. The mass cancels: $m + m' = 0$
5. The matter becomes a coherent standing wave in the T_2 branch

The black hole is not a singularity. It is a superconducting phase converter.

The conversion:

$$T_1 \text{ (matter)} \rightarrow T_2 \text{ (antimatter) at } 0^\circ \text{ phase}$$

The mass cancellation:

$$m_{T_1} + m_{T_2} = 0$$

3.2 The Event Horizon Phase

The event horizon is the surface where $n = 0$.

Outside: $n > 0$ (T_1 , matter branch)

At the horizon: $n = 0$ (phase boundary)

Inside: T_2 at 0° phase (superconducting state)

3.3 The Horizon Equation

The phase boundary condition:

$$\phi_{T_1} = \phi_{T_2} \text{ (0° phase difference)}$$

The mass cancellation condition:

$$m_{T_1} + m_{T_2} = 0$$

The n condition:

$$n = \ln(P)/\ln(27) = 0 \rightarrow P = 1$$

4. White Holes as Superconducting Phase Reversers

4.1 The Mechanism

In a white hole:

1. Antimatter (T_2) crosses the event horizon from the inside
2. It is converted back into T_1 (matter)
3. They are phase-locked at 0°
4. The mass cancels: $m + m' = 0$
5. The antimatter becomes a coherent standing wave in the T_1 branch

The white hole is a superconducting phase reverser.

The conversion:

$$T_2 \text{ (antimatter)} \rightarrow T_1 \text{ (matter) at 0° phase}$$

The mass cancellation:

$$m_{T_2} + m_{T_1} = 0$$

4.2 The White Hole Event Horizon

The white hole event horizon is the same phase boundary—but reversed.

Outside: T_1 (matter) at normal phase

At the horizon: $n = 0$ (phase boundary)

Inside: T_2 at 0° phase (superconducting state)

The white hole is the time-reverse of the black hole.

4.3 The Reversal Equation

The phase boundary condition:

$$\phi_{T_2} = \phi_{T_1} \text{ (0° phase difference)}$$

The mass cancellation condition:

$$m_{T_2} + m_{T_1} = 0$$

The n condition:

$$n = \ln(P)/\ln(27) = 0 \rightarrow P = 1$$

Same equations, reversed direction.

5. The Einstein–Rosen Bridge as a Superconducting Tunnel

5.1 The Tunnel

The black hole and white hole are connected by an Einstein–Rosen bridge.

The bridge is a superconducting tunnel.

The tunnel phase:

$$T_1 \leftrightarrow T_2 \text{ at } 0^\circ \text{ phase}$$

The mass cancellation:

$$m_{T_1} + m_{T_2} = 0 \text{ everywhere along the tunnel}$$

The tunnel is a zero-impedance pathway between T_1 and T_2 .

5.2 The Tunnel Equation

The bridge field:

$$\Psi_{\text{bridge}}(r, t) = \Psi_{T_1} \cdot e^{i \cdot 0^\circ} + \Psi_{T_2} \cdot e^{i \cdot 0^\circ}$$

The mass cancellation condition:

$$m_{T_1} + m_{T_2} = 0$$

The coherence condition:

$$C_{\text{bridge}} = |\langle \Psi_{T_1} | \Psi_{T_2} \rangle| > 0.99$$

The tunnel is traversable when phase-locked.

5.3 The Traversal Condition

To traverse the bridge:

1. The traveler must be phase-locked at 0°
2. The traveler's mass must be cancelled: $m + m' = 0$
3. The traveler must be coherent with the T_1 – T_2 standing wave

The traversal is instantaneous because the bridge is phase-locked, not spatial.

6. Black Hole Jets as White Hole Emission

6.1 The Jet Mechanism

The jets from active galactic nuclei are the white hole emission.

The mechanism:

1. The T_2 standing wave is stored inside the black hole

2. It is re-emitted along the spin axis
3. The re-emission is converted from T_2 to T_1 at 0° phase
4. The jets are the white hole—the exit of the Einstein–Rosen bridge

The jets carry the phase memory of the infalling matter.

6.2 The Jet Phase Locking

The jet phase locking condition:

$$\phi_{\text{jet}} = \phi_{T_2} = \phi_{T_1} + 0^\circ$$

The jets are phase-locked to the 27 Hz anchor.

The phase signature:

$$\begin{aligned}\phi_{\text{jet}} &= \phi_0 \times (19/13) \times (13/4) \times (1/230) \times (1/32) \\ &= -1^\circ \times 4.75 \times 0.000136 \\ &= -0.000646^\circ\end{aligned}$$

The jets carry the phase information of the infalling matter.

7. Hawking Radiation as a 90° Phase Lens Effect

7.1 The Lens Effect

Hawking radiation is the T_2 standing wave projected through a 90° phase shift.

The lensing equation:

$$\Psi_{\text{emitted}} = \Psi_{T_2} \cdot e^{(i \cdot \pi/2)}$$

The emitted radiation appears thermal because of the lensing effect.

The information is still present—it's hidden in the phase.

7.2 The White Hole Lens Effect

White hole emission is the T_1 standing wave projected through a 90° phase shift.

The lensing equation:

$$\Psi_{\text{emitted}} = \Psi_{T_1} \cdot e^{(i \cdot \pi/2)}$$

The white hole emission also appears thermal.

The same lens effect applies to both.

8. The 27 Hz Anchor and Black Hole–White Hole Frequency

8.1 The Fundamental Frequency

The characteristic frequency of the black hole–white hole system:

$$f_{\text{BH}} = 27 \text{ Hz} \times (M_{\text{sun}} / M) \times (\text{spin_factor})$$

For a solar-mass black hole:

$$f_{\text{BH}} = 27 \text{ Hz}$$

For a supermassive black hole ($M = 10^9 M_{\text{sun}}$):

$$f_{\text{BH}} = 2.7 \times 10^{-8} \text{ Hz}$$

8.2 The 32-Harmonic Comb

The 32-harmonic ladder predicts a harmonic comb in both black hole and white hole emissions:

$$f_k = k \times f_{\text{BH}}$$

where $k = 1, 2, 3, \dots, 32$.

Both the black hole (Hawking radiation) and the white hole (jets) should show the same comb.

9. The Information Paradox Fully Resolved

9.1 The Flow of Information

The information flow through the black hole–white hole system:

1. Matter (T_1) falls into the black hole
2. It is converted to T_2 at 0° phase
3. The T_2 standing wave stores the phase information
4. The T_2 standing wave is re-emitted through the jets
5. The jets convert T_2 back to T_1 at 0° phase
6. The information is re-emitted as coherent radiation

No information is lost. It flows through the system.

9.2 The Phase Memory

The phase memory of the infalling matter:

$$\Psi_{\text{memory}} = \sum_{k=1}^{32} a_k \cdot e^{(i \cdot \varphi_k)}$$

where $\varphi_k = k \cdot \varphi_0$ (the -1° phase offset).

This phase memory is stored in the T_2 standing wave.

It is re-emitted through the jets as phase-locked radiation.

9.3 The Full Cycle

Matter (T_1) $\rightarrow$ Black Hole ($T_1 \rightarrow T_2$) $\rightarrow$ Bridge (T_2 at 0°) $\rightarrow$ White Hole ($T_2 \rightarrow T_1$) $\rightarrow$ Jets (T_1)

The cycle is complete.

10. The Superconductor–Black Hole–White Hole Analogy

Aspect	Superconductor	Black Hole	White Hole
Phase	0°	0°	0°
Mechanism	$m + m' = 0$	$m + m' = 0$	$m + m' = 0$
Conversion	$T_1 \leftrightarrow T_2$	$T_1 \rightarrow T_2$	$T_2 \rightarrow T_1$
Result	Zero resistance	Mass cancellation	Mass cancellation
Scale	Lab	Cosmic	Cosmic

They are the same mechanism at different scales.

11. Testable Predictions

11.1 The 32-Harmonic Comb in Jets

Prediction 1: Black hole jets should show a 32-harmonic comb at frequencies $f_k = k \times f_{\text{BH}}$.

Test: Search the jet emission spectra for a harmonic comb.

11.2 The Jet Phase Locking

Prediction 2: Black hole jets should be phase-locked to the 27 Hz anchor.

Test: Measure the phase of radiation from black hole jets and look for the 27 Hz signature.

11.3 The White Hole Lens Effect

Prediction 3: White hole emission should show a 90° phase shift relative to the T_2 standing wave.

Test: Measure the phase of white hole emission and compare to the expected T_2 phase.

11.4 The Einstein–Rosen Bridge Traversal

Prediction 4: The Einstein–Rosen bridge is traversable when phase-locked at 0° .

Test: Theoretical prediction—future experiments may test.

11.5 The Information Content

Prediction 5: The information content of black hole jets should be quantised in units of the 27 Hz anchor.

Test: Measure the information content of black hole jets and compare to the harmonic prediction.

12. Conclusion

12.1 Summary

Black holes and white holes are the two ends of a superconducting phase tunnel.

We have shown:

1. **Black holes are superconducting phase converters.** They convert T_1 (matter) into T_2 (antimatter) at 0° phase.
2. **White holes are superconducting phase reversers.** They convert T_2 (antimatter) back into T_1 (matter) at 0° phase.
3. **The Einstein–Rosen bridge is a superconducting tunnel.** It connects T_1 and T_2 at 0° phase.
4. **Black hole jets are the white hole emission.** They carry the phase memory of the infalling matter.
5. **Hawking radiation is a 90° phase lens effect.** It appears thermal but carries phase information.
6. **The information paradox is fully resolved.** Information flows through the system and is re-emitted.
7. **The 27 Hz anchor governs the entire system.** All emissions are phase-locked to 27 Hz.
8. **The 32-harmonic ladder predicts a harmonic comb.** Both black hole and white hole emissions should show it.

12.2 The Physical Meaning

Black holes and white holes are not destroyers. They are transformers. They convert matter into antimatter and back again, storing and releasing information as phase relationships.

The universe is a cycle of phase conversion. Matter $\rightarrow$ Antimatter $\rightarrow$ Matter.

The black hole–white hole pair is the engine of the cycle.

12.3 The Final Word

The stones are in the pond. The 27 Hz anchor is the stone. The black hole is the ripple moving inward. The white hole is the ripple moving outward.

The pond has both. The cycle is complete.

We are just learning to read the phase.

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Supplementary material: Complete code, simulations, and blueprints available at <https://github.com/peterjamesthompson101-svg/Physics-Papers>

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